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Evaluation of Four Sampling Techniques for Surveillance of *Culex quinquefasciatus* (Diptera: Culicidae) and Other Mosquitoes in African Rice Agroecosystems

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ABSTRACT Field studies were conducted in a rice, *Oryza sativa* L., agroecosystem in Mwea Kenya to compare the efficiency of CO_2-baited Centers for Disease Control (CDC) light traps against nonbaited CDC light traps and gravid traps against oviposition traps in outdoor collection of *Culex quinquefasciatus* Say (Diptera: Culicidae) and other mosquitoes. Collectively, 21 mosquito species from the genera *Culex*, *Anopheles*, *Mansonia*, *Ficalbia*, and *Aedes* were captured during the 10-wk study period. *Cx. quinquefasciatus* was the predominant species in all trap types with proportions ranging from 57% in the nonbaited CDC light traps to 95% in the gravid traps. Significantly higher numbers of *Cx. quinquefasciatus* and *Culex annulioris* Theobald were collected in the CO_2-baited CDC light traps than in the nonbaited CDC light traps, but the numbers of other mosquito species, including malaria vectors *Anopheles arabiensis* Patton and *Anopheles funestus* Giles did not differ significantly between the two trap types. More *Cx. quinquefasciatus* females were collected in grass infusion-baited gravid traps than egg rafts of this species in oviposition traps containing the same infusion. Although most mosquitoes captured in CO_2-baited and nonbaited CDC light traps were unfed, most of those collected in gravid traps were gravid. From these findings, it is concluded that at least in the rice-growing area of Mwea Kenya, CO_2-baited CDC light traps in conjunction with gravid traps can be used in monitoring of *Cx. quinquefasciatus* both for control and disease surveillance.

KEY WORDS *Culex quinquefasciatus*, trapping systems, rice agroecosystems, Kenya

Culex quinquefasciatus Say (Diptera: Culicidae) is a cosmopolitan mosquito species occurring in a wide range of larval habitats, including vegetated lakes and river margins, natural ponds, rice fields, agricultural ponds, waste water treatments, and septic tanks. Its ability to use a myriad of aquatic habitats for larval development has favored its adaptability in both rural and urban environments (Wijers and Kiilu 1977, Beier et al. 1990, Muturi et al. 2006). In Africa, *Cx. quinquefasciatus* is a major vector of Bancroftian filariasis, a potential vector of arboviruses, and an important pest species. Previous studies indicated *Cx. quinquefasciatus* to be an important vector of Bancroftian filariasis only in urban environments (Wijers and Kiilu 1977), but subsequent studies have demonstrated its importance in rural settings particularly where irrigated agriculture is practiced (Udonsi 1988, Thompson et al. 1996, Mwandawiro et al. 1997). Its role as a vector and nuisance species has made it an important target for surveillance, both for monitoring and disease surveys as well as for control efforts.

Mosquito surveillance and monitoring requires accurate sampling techniques based on the behavior and ecology of the target species. In Africa, pyrethrum spray-catches (PSCs), human landing collections, and conventional light traps have been the mainstay of adult mosquito sampling (Lines et al. 1991, Mbogo et al. 1993, Mathenge et al. 2005). However, each of these techniques has its own drawbacks that interfere with its effectiveness. Pyrethrum spray-catches collect indoor resting mosquitoes and are likely to exclude or underestimate exophilic species. In addition, they may underestimate filarial infectivity rates in mosquitoes, because infective larvae are lost during feeding (McMahon et al. 1981), and they also infringe on peoples' privacy. Human landing collection is disfavored because of its labor-intensiveness; dependence on a wide range of environmental, site, and collector factors, and the risk of exposing the collectors to pathogen transmission (Reiter and Gubler 1997). Conventional traps such as light traps are moderately effective for pop-

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ulation sampling, but they underrepresent certain *Culex* species (Tsai et al. 1988, Nayar et al. 2001). Moreover, most mosquitoes collected by conventional traps are usually unfed, and this may greatly interfere with arbovirus surveillance, because unfed mosquitoes are less likely than blood-fed or gravid females to contain virus.

Kairomone-baited light traps and gravid and oviposition traps offer alternatives to the above-mentioned techniques for sampling of *Culex* mosquitoes, and they have been incorporated into mosquito surveillance systems in developed countries. Addition of attractants such as CO_2 and octenol in light traps have been found effective in collection of several mosquito species (Canyon and Hii 1997, Mboera et al. 2000a, McCardle 2004), whereas the use of gravid traps in conjunction with baited-light traps have been found to increase the recovery rate for arboviral infections (Tsai et al. 1989, Nayar et al. 2001, Nasci et al. 2002). Gravid traps are designed to collect gravid female mosquitoes that are attracted to a potential oviposition site containing water high in organic matter (Reiter 1983, 1987). Their ability to collect up to 95% of gravid females (Reiter et al. 1986) enhances arboviral detection in field surveys (Reisen and Meyer 1990). Oviposition traps are used to monitor the abundance and seasonal distribution of *Culex* mosquitoes by measuring oviposition activity (Lee and Rowley 2000).

Despite the usefulness of these techniques in mosquito surveillance, little is known about their effectiveness in Africa. In Tanzania, gravid traps baited with either pheromone or grass infusion were more effective in collection of gravid *Cx. quinquefasciatus* than traps baited with tap water (Mboera et al. 2000b), whereas CO_2-baited Centers for Disease Control (CDC) light traps were superior to nonbaited CDC light traps in collection of the same species (Mboera et al. 2000a). The current study aimed at comparing the performance of gravid traps against oviposition traps and conventional CDC light traps against CO_2-baited CDC light traps for sampling *Cx. quinquefasciatus* in a rice, *Oryza sativa* L., agroecosystem in central, Kenya.

Materials and Methods

Study Area. The study was conducted in Mwea division in Kirinyaga District, 100 km northeast of Nairobi. The study area has been described previously (Mutero et al. 2004). Mwea rice scheme occupies the lower altitude zone of Kirinyaga District in an expansive low-lying area characterized by black cotton soil. The annual rainfall varies from a maximum of 1,626 mm to a minimum of 356 mm, with an average of 950 mm per year. The average temperatures are 21.3°C (range 16.0–26.5°C) and the relative humidity averages 59.5% (range 52–67%). According to the 1999 national census, Mwea division has an estimated 150,000 persons in 25,000 households. The Mwea Irrigation Scheme is located in the West Central region of Mwea division and covers an area of ≈13,640 ha. More than 50% of the scheme area is used for irrigated rice cultivation, whereas the remaining area is used for subsistence farming, grazing, and community activities. The species of the genus *Culex* that have been reported as adults in the area include *Cx. quinquefasciatus*, *Culex poicilipes* Theobald, *Culex annulioris* Theobald, and *Culex tigripes* De Grandpre and De Charmoy, and their abundance is related to the rice cropping cycle (Muturi et al. 2006). The presence of potential vectors of Bancroftian filariasis and arboviruses in the area makes it an important target for vector control studies. *Anopheles arabiensis* is the main malaria vector and the only sibling species of *An. gambiae* s.l. in the area (Mutero et al. 2000, 2004).

Two villages within the irrigation scheme, Kangichiri and Kiuria, were selected for the study. The two villages are 500 m apart and are located at the west central region of the scheme and have ≈150 and 222 homesteads, respectively, with >650 residents in each village. Cows, goats, chickens, and donkeys are the primary domestic animals in the two villages. These animals are kept within 5 m of most houses. More than 90% of the houses have mud walls with iron roofing. More than 75% of each village land is under rice cultivation, whereas people occupy the remaining area with <10% used for vegetables and bananas. Water distribution in the two villages is managed by the National Irrigation Board, and farmers in these villages follow a definite rice cropping program as determined by the board. Many species of birds and mammals visit the rice fields for feeding from the surrounding areas. These can act as intermediate hosts for arboviruses.

Comparison of CO_2-Baited CDC Light Traps with Nonbaited CDC Light Traps. The differences in mosquito species composition and abundance captured in CDC light traps (J.W. Hock Ltd., Gainesville, FL) baited with CO_2 derived from dry ice were compared with conventional CDC light traps. For the CO_2-baited traps, ≈2.0-kg pieces of dry ice were placed in padded envelopes and hung directly above the CDC light trap. In each of the two villages, five paired sets of these traps were run twice per week from 1800 to 0700 hours over a 10-wk period between June and August 2006. For all comparisons, the traps were placed ≈10 m apart, and the sets of traps were distributed evenly across the village.

Evaluation of Gravid Traps and Oviposition Traps. The gravid traps consisted of dark-colored pans containing ≈4 liters of grass infusion with a 6-V fan and a collection net (J.W. Hock Ltd.). Oviposition traps, adapted from Reiter (1986) were essentially gravid traps (buckets) without adult collection apparatus so egg rafts rather than adults were collected. In each of the two villages, 10 paired sets of gravid and oviposition traps were run twice per week from 1800 to 0700 hours over a 10-wk study period between June and August 2006. The grass infusion was made by mixing 1.5 kg of fresh grass in 20 liters of tap water in a 25-liter container that was left outside to steep for 3 d. Traps of each type were placed 10 m apart, and the sets of the traps were distributed evenly across the village.

Mosquito Identification. Adult mosquitoes collected in each of the traps were taken to the laboratory where

they were separated from other insects, sorted into subfamily and subsequently identified morphologically to species (Edwards 1941, Gillies and Coetzee 1987). They were further classified physiologically as unfed, blood-fed, half-gravid, or gravid by examining the abdominal appearance. Concurrent with collection of adult mosquitoes from the gravid traps at 0700 hours, oviposition traps were inspected for *Culex* egg rafts and when present, they were preserved in a petri dish lined with moist filter papers. In the laboratory, the egg rafts were hatched individually in larval pans containing tap water, and the resulting first instars fed on Tetramin fish food (Tetra Holding, Inc., Blacksburg, VA., Germany) up to third instar, which were identified morphologically using taxonomic keys (Hopkins 1952). In cases where larval identification was uncertain, larvae were reared to adults for species identifications.

Statistical Analysis. All statistical analyses were performed using SPSS version 11.5 statistical package (SPSS, Inc., Chicago, IL). The differences in the mean number of mosquitoes collected by CO_2-baited and nonbaited CDC light traps were compared using paired sample *t*-tests. Statistical analysis was done after log transformation of adult mosquito counts to normalize the distribution and minimize the standard error. Because the trap collections in the two villages had a common trend, and because the main objective of the study was to compare the effectiveness of the traps and not the differences between sites, data for the two sites were pooled for statistical analysis.

Results

From a total of 14,263 mosquitoes that were collected by CDC light traps during the sampling period, CO_2-baited (n = 9,414) collections were almost twice that of the nonbaited (n = 4,849) collections. The mean number of mosquitoes collected in the CO_2-baited CDC light traps was 19.69 ± 2.49 (mean ± SE) and significantly higher than 12.21 ± 1.34 (mean ± SE) for the nonbaited light traps (t = 6.281, df = 99, P = 0.012). Twenty mosquito species from the genera *Culex*, *Anopheles*, *Mansonia*, *Ficalbia*, and *Aedes* were captured in the nonbaited CDC traps (Table 1). *Cx. quinquefasciatus* constituted 56.8% of the total collection in the nonbaited traps followed by *Anopheles arabiensis* Patton (27.2%), *Anopheles funestus* Giles (6.0%), *Anopheles pharoensis* Theobald (3.5%), *Anopheles coustani* Laveran (1.6%), *Culex annulioris* Theobald (1.6%), and *Aedes circumluteolus* Theobald (1.2%). The other species occurred in quantities smaller that 1% of the total collection. One specimen of *Culex* spp. and three for *Aedes* were not identified to species because of loss of morphological characteristics for identification. In the CO_2-baited CDC light traps, 21 mosquito species were captured (Table 1). The most abundant species in these collections was *Cx. quinquefasciatus* which constituted 72.0% of the total catch, followed by *An. arabiensis* at 14.1%, *An. funestus* at 5.6%, *Cx. annulioris* at 3.2%, and *An. pharoensis* at 2.4%. The other species were represented in quantities lower than 1% of the total capture. Two specimens of *Culex* spp. and one specimen for *Aedes* were unidentifiable.

Table 1. Total number (percentage) of diverse mosquito species caught in CO_2-baited and nonbaited CDC light traps in Kangichiri and Kiuria, Mwea Kenya

Mosquito species	CO_2-baited CDC light traps	Nonbaited CDC light traps
Ae. circumluteolus	58 (0.62)	56 (1.15)
Ae. cumminsi	3 (0.03)	11 (0.23)
Ae. furcifer-taylori[a]	5 (0.05)	4 (0.08)
Aedes sp.	1 (0.01)	3 (0.06)
An. coustani	73 (0.78)	79 (1.63)
An. funestus	526 (5.59)	289 (5.96)
An. arabiensis	1,329 (14.12)	1,318 (27.18)
An. maculipalpis	3 (0.03)	10 (0.21)
An. nili	1 (0.01)	1 (0.02)
An. pharoensis	221 (2.35)	171 (3.53)
An. pretoriensis	1 (0.01)	1 (0.02)
An. rufipes	1 (0.01)	0 (0.00)
Culex sp.	2 (0.02)	1 (0.02)
Cx. annulioris	302 (3.21)	78 (1.61)
Cx. duttoni	8 (0.08)	6 (0.12)
Cx. poicilipes	47 (0.50)	9 (0.19)
Cx. tigripes	1 (0.01)	4 (0.08)
Cx. quinquefasciatus	6,781 (72.03)	2,755 (56.82)
Ficalbia plumosa	3 (0.03)	4 (0.08)
Mansonia africana	46 (0.49)	39 (0.80)
Mansonia uniformis	2 (0.02)	10 (0.21)
Total	9,414	4,849

[a] *Ae. furcifer* and *Ae. taylori* are morphologically indistinguishable and hence treated as *Ae. furcifer-taylori.*

CO_2-baited CDC light traps were more superior to nonbaited CDC light traps in sampling of *Cx. quinquefasciatus* and *Cx. annulioris.* The mean number of *Cx. quinquefasciatus* collected over the sampling period was 71.38 ± 10.79 (mean ± SE) in the CO_2-baited light traps compared with 32.80 ± 5.18 in the nonbaited light traps (t = 9.581, df = 99, P = 0.002). Similarly, the mean number of *Cx. annulioris* collected over the sampling period in the CO_2-baited light traps was 4.79 ± 0.58 (mean ± SE) and significantly higher than 2.17 ± 0.28 (mean ± SE) in the nonbaited traps (t = 10.959, df = 99, P = 0.001). The mean numbers of the other species collected did not vary significantly between the two sampling methods.

In total, 24,308 mosquitoes from four genera and 12 species and a further 428 egg rafts belonging to three species of *Culex* were collected by gravid traps and oviposition traps, respectively (Table 2). *Cx. quinquefasciatus* made up 95.1% of the total collection in gravid traps followed by *An. arabiensis* at 3.5% and *An. funestus* at 1.1%. The other species were represented in quantities lower than 1% of the total capture. In the oviposition trap collections, *Cx. quinquefasciatus* accounted for 92.5% of the total number of egg rafts collected, whereas the remainder were *Cx. duttoni* (6.3%) and *Cx. tigripes* (1.2%).

Most mosquitoes captured in both CO_2-baited and nonbaited CDC light traps were unfed with proportions ranging from 66% for *An. coustani* to 93% for *Cx. annulioris* and *An. arabiensis.* Most mosquitoes collected in the gravid traps were gravid, ranging from

Table 2. Total number of adult mosquitoes (percentage) caught in gravid traps and egg rafts collected in oviposition traps in Kangichiri and Kiuria, Mwea Kenya

Mosquito species	Gravid traps	Oviposition traps
Ae. aegypti	3 (0.01)	—[a]
Ae. furcifer-taylori[b]	3 (0.01)	—
An. coustani	2 (0.01)	—
An. funestus	262 (1.08)	—
An. arabiensis	861 (3.54)	—
An. maculipalpis	1 (0.00)	—
An. pharoensis	11 (0.05)	—
Cx. annulioris	18 (0.07)	0 (0.00)
Cx. duttoni	1 (0.00)	27 (6.31)
Cx. quinquefasciatus	23,122 (95.12)	396 (92.52)
Cx. tigripes	16 (0.07)	5 (1.17)
Ficalbia plumose	8 (0.03)	0 (0.00)
Total	24,308	428

[a] Dash indicates mosquitoes lay eggs singly, and ovitraps were not inspected for their eggs.
[b] *Ae. furcifer* and *Ae. taylori* are morphologically indistinguishable and hence treated as *Ae. furcifer-taylori.*

25% for *An. funestus* and *An. pharoensis* to 89% for *Cx. quinquefasciatus* (Table 3).

Discussion

Carbon dioxide is one of many volatiles produced by humans, reported to elicit behavioral responses by mosquitoes (Gillies 1980). CDC light traps are commonly augmented with CO_2 from dry ice to enhance mosquito capture. In Africa, however, this method has been applied infrequently for the study of outdoor host-seeking behavior of vector mosquitoes (Mboera et al. 2000a). Mboera and Takken (1997) reported that traps using kairomones (olfactory cues used to locate hosts or prey) provide a more objective monitoring tool for the host-seeking fraction of mosquitoes in an outdoor situation where the effect of "competitive" natural human odors is scanty. The current study demonstrated that, although CO_2-baited CDC light traps were not superior to nonbaited CDC light traps in numbers of mosquito species collected, they increased the numbers captured of *Cx. quinquefasciatus* and *Cx. annulioris* but had no effect on the numbers of other mosquito species. Previous studies have reported an increase in the number of *Cx. quinquefasciatus* captured after addition of CO_2 in CDC light traps (Mboera et al. 2000a). These studies further reported a significant decrease in the number of *Cx. quinquefasciatus* collected in CO_2-baited CDC traps when the light was on, leading to the hypothesis that this species is primarily attracted to CO_2 but may become disoriented by the presence of the light, reducing the catch. In the current study, we used CO_2-baited CDC traps in the presence of light and could not, therefore, test this hypothesis. We suggest the need for further studies to assess the validity of this hypothesis in African rice agroecosystems. The lack of significant difference between numbers of *An. arabiensis* and *An. funestus* in the CO_2-baited and nonbaited CDC traps indicates the need to explore other alternative cues that may be attractive to these species outdoors. Under laboratory conditions using *An. gambiae* s.s., Njiru et al. (2006) demonstrated that CO_2 synergizes responses to other human exudates, such as foot odor. The potential of this synergy to enhance collections of *An. arabiensis* and *An. funestus* in rice land agroecosystems should be tested. More importantly, efforts should be made to improve the delivery system for carbon dioxide because the use of this compound in gaseous form or as dry ice remains a major obstacle for its inclusion in trapping devices in the tropics. Recent studies have evaluated the potential of granular sachets and yeast-generated CO_2 as alternatives to dry ice (Oli et al. 2005, Rapaport et al. 2005). The results have shown that, although both methods are inferior to dry ice for mosquito collections, they are cheap, readily available, and may enhance monitoring of mosquitoes in areas where dry ice is not available.

Table 3. Common mosquito species collected by three trapping systems in Kangichiri and Kiuria, Mwea, Kenya, and the numbers (percentage) in different physiological states

Method	Mosquito species	Physiological status				Total
		Unfed	Bloodfed	Half gravid	Gravid	
Baited	*Cx. quinquefasciatus*	6,129 (90.38)	362 (5.34)	56 (0.83)	234 (3.45)	6,781
	Cx. annulioris	282 (93.38)	10 (3.31)	6 (1.99)	4 (1.32)	302
	An. arabiensis	1,205 (90.67)	72 (5.42)	21 (1.58)	31 (2.33)	1,329
	An. funestus	437 (83.08)	64 (12.17)	4 (0.76)	21 (3.99)	526
	An. pharoensis	166 (75.11)	25 (11.31)	21 (9.50)	9 (4.07)	221
	An. coustani	48 (65.75)	15 (20.55)	5 (6.85)	5 (6.85)	73
Nonbaited	*Cx. quinquefasciatus*	2,361 (85.70)	263 (9.55)	16 (0.58)	115 (4.17)	2,755
	Cx. annulioris	72 (92.31)	4 (5.13)	2 (2.56)	0 (0.0)	78
	An. arabiensis	1,232 (93.47)	56 (4.25)	21 (1.59)	9 (0.68)	1,318
	An. funestus	246 (85.12)	36 (12.46)	1 (0.35)	6 (2.08)	289
	An. pharoensis	140 (81.87)	13 (7.60)	9 (5.26)	9 (5.26)	171
	An. coustani	56 (70.89)	19 (24.05)	0 (0.00)	4 (5.06)	79
Gravid traps	*Cx. quinquefasciatus*	2,044 (8.72)	408 (1.74)	227 (0.97)	20,756 (88.57)	23,435
	Cx. annulioris	10 (55.56)	0 (0.00)	0 (0.00)	8 (44.44)	18
	An. arabiensis	329 (39.31)	193 (23.06)	70 (8.36)	245 (29.27)	837
	An. funestus	75 (29.18)	102 (39.69)	16 (6.23)	64 (24.90)	257
	An. pharoensis	6 (50.00)	1 (8.33)	2 (16.67)	3 (25.00)	12
	An. coustani	1 (50.00)	1 (50.00)	0 (0.00)	0 (0.00)	2

Most mosquitoes captured in both CO_2-baited and nonbaited CDC light traps were unfed, confirming previous observations that this method is less desirable for arboviral surveys. To enhance the probability of detecting arboviral infections in *Culex* mosquitoes, gravid traps are commonly used to collect previously blood-fed females in surveillance programs (Reiter et al. 1986, Reisen and Meyer 1990). Gravid traps are augmented with oviposition traps because some mosquitoes collected by these traps are badly damaged to allow for species identification, whereas egg rafts are easy to collect and first instars are identifiable (Dodge 1966, Madder et al. 1980). Both traps require an attractant to lure mosquitoes, and we used grass infusion that is known to be a good attractant and stimulant for a variety of *Culex* species (Madder et al. 1980, Mboera et al. 2000b). In the current study, higher numbers of gravid mosquitoes than egg rafts were collected. Because the odors from the grass infusion are perceived by olfaction, the results suggest that the number of egg rafts deposited may not reflect the number of gravid females visiting an oviposition site (Mboera et al. 2000b). For oviposition activity to occur, an infusion used as an attractant in a trap must be both an effective attractant to lure gravid females and a stimulant to elicit oviposition (Jackson et al. 2005). Gravid traps collect mosquitoes responding to an attractant but do not require oviposition to occur. Another difference between the gravid and oviposition traps was the color of the containers. We used dark pans for gravid traps and white buckets for oviposition traps, and some studies have shown that many mosquito species, including *Cx. quinquefasciatus*, prefer to oviposit in darker containers (Allan and Kline 2004). We recommend the need for a more detailed study to compare the relative attractiveness of several commonly used infusions to collect *Cx. quinquefasciatus* in African rice agroecosystems by using both gravid traps and oviposition traps (Reiter 1986).

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